

Using Real-Time Public Sentiment to Predict Stock Metrics

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Abstract—This paper presents a real-time system that automatically collects \$TSLA stock data from Yahoo Finance and scrapes tweets from X.com, formerly known as Twitter every 10 minutes. Tweets are filtered and preprocessed by removing noise such as special characters, URLs, and mentions. The system combines financial features— Open, High, Low, Close, Volume (OHLCV)—with tweet engagement metrics, including number of views, likes, replies, and retweets. Sentiment is extracted using an NLP-based emotion classifier that assigns each tweet a score indicating its relevance to 11 distinct emotion categories. Two Long Short-Term Memory (LSTM) models are trained: one using only financial data, and the other incorporating both financial and sentiment features. The performance of each model is compared to evaluate the impact of social sentiment on short-term stock price prediction. A web-based dashboard visualizes predictions updated in real time. The research and testing conducted showed that both models are successful in capturing predictions, while the financial only model provided more stable precision – the sentiment enhanced model was able to outperform during certain market conditions like volatile sessions and trading sessions with increased social media chatter relating to the individual stock. Moreover, an optimized model that implemented both prediction architectures and a signal system to determine which prediction model to follow based on market trends and conditions would increase predictive precision while maintaining relative stable performance in normal market conditions.

Index Terms—Real-time system, Yahoo Finance, X.com, Twitter, Sentiment, Natural Language Processing (NLP), Long Short-Term Memory (LSTM), Stock prediction, OHLCV

I. INTRODUCTION

Sentiment refers to the view or attitude towards an event or situation useful in many applications. In today’s digital world, sentiment can be captured from various social medias, news articles, and forums. One area where people’s opinions play a vital role is the stock market, where the public’s perception can influence investors’ investment decisions, causing the price to fluctuate.

Given the volatile nature of the stock market, collecting both financial and sentiment data in real time is important for accurate predictions. Web Scraping combined with Natural Learning Processing (NLP) techniques, can be used to collect and extract sentimental data from textual sources.

Since both sentiment and financial data are inherently time-dependent and sequential, deep learning algorithms are well suited for this task. Recurrent Neural Networks (RNNs),

specifically Long Short-Term Memory (LSTM) networks, are capable of capturing temporal patterns and learning from past sequences to make informed predictions about future stock prices.

The idea of combining sentiment with financial data to predict stock prices is not new; however, previous approaches often limited sentiment to only a few categories and failed to incorporate additional engagement metrics such as the number of views, likes, replies, and retweets that can reflect the level of public agreement or impact.

II. LITERATURE REVIEW

There are numerous learning models and approaches developed over the years to analyze financial trends, predict stock market behavior, and extract sentiment from social media data. In a 2023 study, researchers extracted the OHLC and polarity of news articles listed on the Indonesia Stock Exchange (IDX) using Naive Bayes Classifier for \$BBCA, \$BUMI, and \$ASII [1]. Instead of directly combining the financial and sentiment directly, OHLC was converted to positive, negative, or equal by comparing it to the closing price of the previous days [1]. With the combined categorical data based on date, KNN was used to predict whether the price would rise, fall or remain stable giving a high accuracy of above 90% for all three tickers [1].

The next approaches utilizes neural networks for their ability to learn from mistakes and adapt to data and predict a numerical value. In 2018, a research group used Yahoo Finance data along with articles from the New York Times covering a span of 10 years [2]. The articles were filtered to focus on specific categories such as business, national, world, U.S., politics, opinion, technology, science, health, and foreign affairs, resulting in a selection of 400,000 articles from an initial pool of 1 million [2]. Instead of relying on 1 sentiment predictor, VADER and SentiMo were both used and a voting system was in place to determine the final sentiment scores for mixed, positive, negative, and neutral [2]. Four different training models were tested resulting with Levenberg-Marquardt backpropagation performing the best [2]. 2 models were created, one with only financial data and the other with both financial and sentiment data to compare which performed

better [2]. It was found that sentiment did contribute to the model's accuracy given the influence was tuned down [2].

The next 2 approaches use LSTMs which is more aligned with our proposal. In a paper from 2021, a team used OHLVC from Yahoo Finance to predict Apple stock and used an LSTM to predict whether the tweets from Twitter was bullish or bearish [3]. Before getting the sentiment index, stop words, special characters, addresses, websites, and mentions were removed [3]. The study reported a recall, precision, and F1-score of 0.74 when using an LSTM with attention [3].

Lastly, researchers in 2023 implemented a real-time system to predict the price of Twitter and Facebook using an NLP for sentiment scores from web crawling and an LSTM for prediction [4]. Like the previous approach, the data was preprocessed by removing special characters, URLs, and converting to lowercase [4]. Their model was overfitting at the 40th epoch and had an 80% accuracy with precision, recall, and f1-score at 79% [4].

III. METHODOLOGY

Similar to [4], we propose a real-time system that will collect finance data from Yahoo Finance for \$TSLA and web scrape X.com, formally known as Twitter, every 10 minutes shown in Fig. 1. Instead of the primary 3 categories, positive, negative, and neutral, used in [1]–[4], there will be 11 categories shown in Section IV as well as the additional engagement metrics mentioned in Section I. Using inspiration from [3], [4], some tweets will be filtered out and preprocessing will clean the text before passing to the NLP transformer shown in Section V. After preprocessing, two LSTMs will be tested, one with financial data only, and the other with sentiment and financial data similar to [2] explained in Section VI. A local hosted website is also provided showing a line graph of predictions 10 minutes ahead updated in real time. Full implementation of the real-time system application can be found in [5].

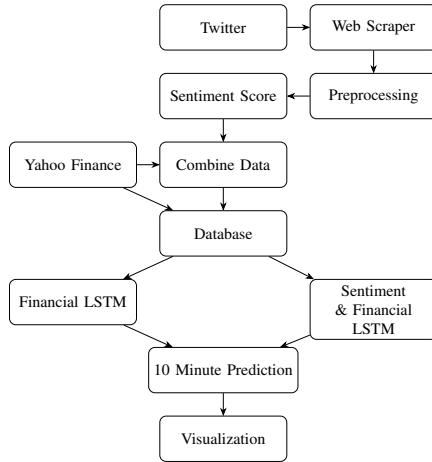


Fig. 1. Overview of the proposed real-time sentiment & stock prediction system.

IV. COLLECTION OF DATA

The financial data is sourced from Yahoo Finance using the unofficial API called yfinance. Data is fetched every 10 minutes with a 1-minute interval capturing the OHLVC metrics: Open, High, Low, Close, and Volume. Open is the start price while close, which is the target for prediction, is the ending price within the minute. The high and low reveal the price extremes in the minute, showing volatility and potential resistance or support levels. Lastly, the volume indicates the level of trading activity, offering insight into market interest and momentum. The data is then stored onto a local SQLite file in its own table specifically for the finance only model and will be referred to after sentiment assignment.

Alongside the financial data, English tweets containing the tag \$TSLA were web scraped from Twitter's latest timeline every 10 minutes using a search query. While 10 minutes seems long for a volatile system, it is necessary to avoid rate limits and bot detection. Not all tweets get selected to avoid junk data. For instance, tweets containing Discord or Telegram links are ignored as most of the time people include a bunch of popular tickers to advertise their service or server. Similarly, tweets that have 10 or more consecutive tickers are skipped for the same reasons. On average, the program would collect approximately 20 tweets per batch. From the HTML, the unique ID, text, reply count, view count, like count, retweet count, and creation date was collected.

After preprocessing, which is explained in Section V, the text was processed using a pretrained RoBERTa NLP transformer model, which was specifically fine-tuned on Twitter data. When passed, the transformer gives the percentage of how closely it matches each category and will be appended to the tweet. The 11 categories include anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, and trust.

After the sentiment scores, the data are combined with the financial data based on the creation date to the truncated minute. The data will be stored in its own table specifically for the financial and sentiment LSTM. The data is then passed into both LSTMs for prediction.

V. PREPROCESSING

Preprocessing the text from the tweets is an essential step for standardization and normalization, ensuring that the data is clean, consistent, and suitable for input into the RoBERTa model. The first step is to convert emojis into text format as they carry contextual meaning just like regular text. Additionally, user mentions are replaced with their corresponding names. For instance, "@elonmusk" is replaced with "Elon Musk" as his name is often brought up since he is the CEO of Tesla. Other mentions are generalized to "[Mention]" to avoid introducing bias or skewed results, especially when usernames contain emotionally charged or sentiment-laden words. Finally, line breaks and any extra spaces are removed due to the structure of Twitter's HTML and the nature of web scraping.

VI. LSTM MODEL'S ARCHITECTURE

A. Financial Data Only

Our initial LSTM model predicts Tesla stock prices 10 minutes ahead using historical OHLCV data. The model processes 36-minute sequences to capture relevant temporal patterns for prediction.

The architecture consists of:

- 1) An input layer accepting sequences of shape (36, 5), where 5 represents the OHLCV features.
- 2) A first LSTM layer with 160 units returning sequences to the next layer, followed by a dropout layer (rate = 0.2).
- 3) A second LSTM layer with 120 units outputting only the final state, followed by another dropout layer.
- 4) A dense output layer with a single unit for price prediction.

We use the Adam optimizer (learning rate = 0.0001) with Mean Squared Error as the loss function and track Mean Absolute Error as an additional metric. To prevent overfitting, early stopping monitors validation loss with a patience of 40 epochs. The model trains with a batch size of 32 for up to 150 epochs on data split into training (75%), validation (10%), and testing (15%) sets. All features are normalized using MinMaxScaler fitted only on training data to prevent data leakage.

B. Financial and Sentiment Data Combined

Our enhanced model integrates financial data with sentiment and engagement metrics from social media while maintaining the same sequence length. The model processes three feature categories:

- 1) Financial features: Five OHLCV metrics.
- 2) Sentiment features: Eleven emotional dimensions from tweets (anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, and trust).
- 3) Engagement features: Four tweet interaction metrics (replyCount, viewCount, likeCount, retweetCount).

This approach processes 20 features per time step, with configurable weighting factors for each feature category implemented during preprocessing.

The architecture includes:

- 1) A bidirectional LSTM first layer (160 units) processing sequences in both directions.
- 2) A second LSTM layer (120 units) maintaining sequence information.
- 3) A third LSTM layer (80 units) consolidating temporal information.
- 4) Dropout layers (rate = 0.2) after each LSTM layer.
- 5) A dense output layer for price prediction.

The model uses the same optimizer, loss function, and training parameters as the financial-only model. The preprocessing handles NaN values and applies separate scaling for features and targets.

This architecture integrates public sentiment with financial data, aiming to capture additional signals that might influence stock price movements beyond traditional metrics.

VII. RESULTS

We tested both models through real-time prediction of Tesla stock prices during April 2025 trading days. The performance was measured using standard metrics and visual comparison.

The financial-only model performed well with a test set RMSE of 0.7126 and R^2 of 0.9666. Root Mean Squared Error (RMSE) of 0.7126 indicates that the model's predictions deviate from actual prices by approximately \$0.71 on average. With Tesla trading around \$292, this represents about 0.24% of the share price. The high coefficient of determination (R^2) of 0.9666 means the model explains nearly 97% of the variance in Tesla's stock price movements. The model maintained a Mean Absolute Error (MAE) of 0.5388, representing an average absolute deviation of \$0.54 between predicted and actual stock prices. The Mean Absolute Percentage Error (MAPE) of 18.96% indicates the model's predictions deviate from actual prices by approximately 19% relative to the true price. This model reached its best validation loss at epoch 92.

The sentiment enhanced model showed similar results with a test RMSE of 0.9774 and R^2 of 0.9413. The RMSE of 0.9774 means this model's predictions deviate from actual prices by about \$0.98 on average. The R^2 value of 0.9413 indicates the model explains about 94% of the price variance. Its MAE of 0.6977 shows predictions differ from actual values by about \$0.70 on average. The MAPE of 24.58% shows predictions deviate from actual prices by about 25% relative to the true price. This model achieved best validation loss at epoch 101.

Analysis of both training metrics and daily predictions revealed distinct performance characteristics between the two approaches. While the financial-only model more consistently followed price movements, it lagged behind the sentiment enhanced model during sudden abnormal price shifts and days where social media activity relating to Tesla stock price was more prevalent.

As mentioned, the sentiment-enhanced model performed particularly well during periods of higher social media activity. For example, on April 22 and April 23, when Tesla received significantly more attention on X.com compared to other days, the sentiment model was able to track price changes more accurately than the basic financial model. This is evident in Fig. 2 and Fig. 3, where the blue line represents the actual closing price, the red line shows the prediction based solely on financial data, and the green line depicts the prediction using both financial and sentiment data. This suggests the sentiment features provided valuable additional data during periods of heightened market discussion allowing for more precise predictions.



Fig. 2. Real-Time Stock Price Prediction Comparison (April 22, 2025)



Fig. 3. Real-Time Stock Price Prediction Comparison (April 23, 2025)

Overall, these results indicate that while the financial-only model provides more consistent performance across various market conditions, the sentiment-enhanced approach offers particular advantages during periods of high social media engagement and market volatility.

VIII. CONCLUSION AND FUTURE WORK

This project investigated the effectiveness of integrating social media sentiment analysis with traditional financial data for predicting short-term stock price movements. We developed two LSTM-based models: one using only financial data and another incorporating sentiment features extracted from social media posts.

Our experiment results demonstrated that while the financial-only model maintained better overall accuracy metrics (RMSE of 0.7126 vs. 0.9774), the sentiment-enhanced model showed superior performance during periods of high social media activity and volatile market conditions. This confirms our hypothesis that integrating sentiment analysis can capture market dynamics not reflected in historical price data alone, particularly during emotionally-driven trading periods.

Several directions for future work emerge from this research. First, developing a hybrid model that dynamically switches between sentiment-enhanced and financial-only predictions based on market conditions could combine the strengths of both approaches. Such a system could monitor social media volume and market volatility in real-time, selecting the optimal prediction model based on current conditions rather than applying a one-size-fits-all approach.

Second, expanding sentiment data sources beyond X.com to include financial news articles, Reddit forums, StockTwits, and other specialized investment communities would provide a more comprehensive view of market sentiment. Different platforms often capture different investor demographics, potentially revealing sentiment patterns not visible from a single source.

Third, exploring temporal relationships between sentiment shifts and price movements could enhance prediction accuracy. Determining when sentiment leads, lags, or coincides with price changes in different market contexts would allow for more nuanced model training.

Finally, practical applications of this research extend beyond trading systems. These models could serve as market monitoring tools for regulatory bodies, risk assessment frameworks for portfolio managers, and sentiment indicators for corporate communications teams evaluating public perception of company announcements. Instead of trying to predict the closing price, it could serve to predict whether prices would rise, fall, or remain stable similar in [1].

As markets continue to be influenced by sentiment-driven trading, especially during periods of uncertainty, models that effectively integrate both quantitative and qualitative factors will become increasingly valuable for understanding price dynamics for Tesla stock and the whole stock market.

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